

# Marginally Useful: Conformal Prediction is Early Stopping

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## Abstract

Split conformal prediction ends where its calibration begins: fit a model, pool its held-out residuals, quote the pooled quantiles, stop. Of 686 arXiv papers titled conformal prediction, 599 stop there. The other 87 replace the pooled law, tune its level, use it downstream, or propose no method. None models past it. We introduce and prove the conformal information gap identity, fixing a conformal scheme’s excess log score before calibration. On 17,364 economic series, stopping costs 0.40 to 0.48 nats after a null-control adjustment, composing past the pooled law recovers a third to a half, and coverage detects none of it.

## 1 Introduction

Conformal prediction attaches a guarantee to any point forecaster by a recipe. Fit a model however you like. On  $n$  held-out points, compute a scalar error for each, for instance the absolute residual. Sort the errors, take the  $\lceil (n+1)0.9 \rceil$ -th smallest, and quote the model’s prediction plus or minus that value. If the held-out data and the future point are exchangeable, meaning their joint law is unchanged by shuffling, the quoted interval contains the truth at least 90% of the time, at any sample size, with no assumption on the model or the data distribution.

The recipe ends with the percentile, and nothing is fitted to the output of the pooled-error step. In machine learning, early stopping is a regularizer chosen because validation loss says further fitting would hurt. We have found no paper that argues for the conformal stop on any such grounds.

We measure log scores in nats. A forecaster whose average log density trails another’s by 0.15 nats assigns the realized outcomes  $e^{0.15} \approx 1.16$  times less probability per observation. Beside it we report CRPS, the continuous ranked probability score, which integrates the squared difference between the forecast’s cumulative distribution and the outcome’s step function. It is in the units of the data, lower is better, and most conformal papers report it.

We make three contributions. We count how often published conformal methods stop at the pooled law (Section 2). We introduce and prove the conformal information gap identity (Proposition 2), which fixes, exactly and at a fixed calibration sample, the part of the excess log score that no calibration choice can reach. We then measure what stopping costs on economic series, and how much of that cost composing a model past the pooled law recovers (Section 4).

**Related work.** The proof of Proposition 2 uses one standard identity of information theory (Lemma 1).

The most any forecast can gain over the unconditional law is the mutual information between outcome and inputs (DelSole, 2004). Catt (2026) takes this quantity as the definition of forecastability, and Corollary 5 shows how much of it a conformal scheme captures.

Three published results concern the pooled shape. Exact distribution-free conditional coverage is impossible in nonatomic settings (Lei and Wasserman, 2014; Barber et al., 2021). Weighting calibration points unequally costs coverage in proportion to the total-variation distance between their residual laws (Barber et al., 2023). A guarantee held at a fixed point rather than on average inflates the threshold for classical testing-power reasons (Li, 1989; Low, 1997; Cai and Low, 2004). Together these give the pooled shape a defence. None of them concerns the stop. The base forecaster is the subject of Cotton (2026).

## 2 How often published methods stop at the pooled law

We classified every arXiv paper with “conformal prediction” in the title as of 20 August 2026, 686 papers. A keyword screen assigned the labels, and we read in full every paper whose label could change the conclusion. The classifier, the frame and the per-paper verdicts are in the repository.

The object classified is the *empirical map*: the step turning the multiset of held-out errors into a distribution, concretely the empirical cumulative distribution of the calibration scores. A paper *stops* if the output of that map, its quantiles, its  $p$ -values, or its intervals, is the answer. A paper *replaces* if it substitutes a conditional estimate for the pooled quantile, as SPCI fits a quantile regression to recent residuals (Xu and Xie, 2023). It *tunes* if it keeps the map and adapts the level online, as adaptive conformal inference does (Gibbs and Candès, 2021). It *consumes* if it feeds the output to a downstream decision such as screening pseudo-labels. It *composes* if it keeps the map and fits a model to its output.

Of 686 papers, 599 stop, 16 replace, 28 tune, 38 consume, and 5 propose no method. One compose candidate was flagged by the screen: distributional conformal prediction (Chernozhukov et al., 2021), which uses an estimated conditional distribution as its score. Read in full, it conditions *before* the map, and its answer is the empirical quantile of the resulting ranks. It uses a richer score and then stops. A paper would count as composing if it kept the empirical rank transform, applied it to residuals, and fitted a downstream model on the transformed stream. Reweighting the empirical law, conditioning upstream and adapting the level do not count.

Because the screen’s default label is *stop*, we also coded a random sample by hand. We drew 100 papers with a fixed seed from a wider archive of conformal prediction papers and labelled them without reference to the screen. We find that 93 of the 100 treat the conformal step as terminal. The remaining 7 are unclear because they pass its output downstream as a filter or a feature. None fits a further law on the transformed scale and carries it back as a forecast.

Outside conformal prediction the composing move is routine. The normal quantile transform in hydrological forecasting (Bogner et al., 2012), copula time-series models (Chen and Fan, 2006), and autoregressive transformation models (Rügamer et al., 2023) all apply an empirical marginal map and keep modelling in the transformed coordinates.

A second classification, in the repository, records how papers pool residuals. In 94.5% of them the calibration residuals are pooled, either all together or within fixed strata, and the weights never vary smoothly with the test point. Nine papers weight smoothly and claim exact validity. We read all nine in full. Each obtains validity by marginal recalibration, a stated coverage gap, a known covariate-shift tilt, or added smoothness assumptions.

### 3 The excess log score fixed before calibration

A *nonconformity score* is a map  $A : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$  assigning a scalar to each candidate outcome  $y$  at input  $x$ . A *stratification*  $\kappa : \mathcal{X} \rightarrow \mathcal{K}$  assigns each input to a category. Both are fit on a training fold and frozen, since exchangeability of calibration and test scores is what the coverage argument uses. The schemes covered by the proposition are those whose score is, for each  $x$ , a strictly increasing absolutely continuous bijection in  $y$ : signed-residual and normalized signed-residual schemes, and predictive systems built on invertible score transformations. Folded scores such as the absolute residual  $|y - \hat{\mu}(x)|$  and quantile-regression scores are not bijections in  $y$ . Remark 7 treats the absolute residual in its symmetric reading. Quantile-regression scores need a separate treatment. The empirical arms in Section 4 are of the covered type.

Such a scheme emits the pooled empirical law of calibration scores in the test point's stratum, pushed back through  $A(x, \cdot)^{-1}$ . For log scoring the emitted law must have a density, so let  $\hat{G}_\kappa$  be an absolutely continuous estimate of the pooled score law (a kernel-smoothed or interpolated empirical law). The raw empirical measure is atomic and would score  $-\infty$ . Write  $W = A(X, Y)$ , let  $G_\kappa$  be the population pooled law, and let the benchmark be the conditional oracle, whose expected negative log score is the conditional entropy  $H(Y | X)$ . Throughout,  $L(\hat{F}) = \mathbb{E}[-\log \hat{f}_X(Y)]$  is the expected log loss of the conformal predictive, so an excess is positive and smaller is better.

The proof of the floor uses one standard identity, the chain rule for relative entropy along a product reference (Cover and Thomas, 2006). In coding theory it splits redundancy into a dependence term and a mismatch term (Clarke and Barron, 1990; Grünwald, 2007).

**Lemma 1** (Chain rule for relative entropy). *Let  $(U, V)$  have a joint density with  $I(U; V) < \infty$ , let  $p_V$  be the marginal density of  $V$ , and let  $q$  be any density with  $\mathbb{E}|\log q(V)| < \infty$ . Then*

$$\mathbb{E} \log \frac{p(V | U)}{q(V)} = I(U; V) + \text{KL}(P_V \| Q).$$

*Proof.* Insert the marginal density  $p_V$  inside the logarithm:

$$\mathbb{E} \log \frac{p(V | U)}{q(V)} = \mathbb{E} \log \frac{p(V | U)}{p_V(V)} + \mathbb{E} \log \frac{p_V(V)}{q(V)}.$$

The first expectation is  $I(U; V)$  by definition. In the second the integrand depends on  $V$  alone, and  $V$  has law  $P_V$ , so it equals

$$\int p_V(v) \log \frac{p_V(v)}{q(v)} dv = \text{KL}(P_V \| Q).$$

Both terms are finite under the assumptions. □

The floor needs two further facts about conformal schemes. The first is that the Jacobian of an invertible score cancels between truth and forecast, so the input reaches the predictive only through the frozen score and stratum. The second is that every calibration operation acts on the pooled scores alone. Together they give the following result, which to our knowledge has not been stated before.

**Proposition 2** (Conformal information gap identity). *Fix the training fold and the calibration sample, so  $(A, \kappa, \hat{G}_\kappa)$  are deterministic, let  $(X, Y)$  be a test pair independent of both, and assume*

the laws below admit densities with finite divergences. Then the excess log loss of the conformal predictor decomposes exactly as

$$L(\hat{F}) - H(Y | X) = I(W; X | \kappa) + \mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa),$$

where  $I(W; X | \kappa)$  is the conditional mutual information between score and input given the stratum. We call it the conformal information gap, or the floor. (Unconditionally, take the outer expectation over the calibration sample. The first term is unchanged.) The second term is nonnegative and is the only part any calibration choice, estimator, quantile convention, randomization, or level, can affect. At a fixed calibration sample it vanishes if and only if  $\hat{G}_\kappa = G_\kappa$  almost everywhere. KL-consistent estimation of the pooled law with growing calibration mass per stratum drives it to zero asymptotically, which is an assumption on the estimator, not a consequence of the setup.

The proof is a change of variables in which the Jacobian of  $A$  cancels between truth and forecast, then Lemma 1 within each stratum (Appendix A). The input reaches the predictive only through the frozen score and the frozen stratum. Whatever information about the score the input still carries within a stratum is therefore lost before calibration begins, and more calibration data does not recover it. With a location score and one stratum the floor is  $I(R; X)$ , the mutual information between the model's error and its input.

*Remark 3* (When the floor is zero). The floor is zero exactly when the frozen pair renders the score independent of the input within strata:  $I(W; X | \kappa) = 0$  iff  $W \perp X | \kappa$ . When  $X$  carries the full information set available downstream, that is the statement that the modelling was finished before the conformal step. Conformal calibration is especially valuable when one is unwilling to assume that. Its guarantee does not diagnose whether the floor is zero, however, and one may deploy it on perfectly finished residuals purely for finite-sample validity. When the floor is positive, no calibration choice recovers any of it.

**Lemma 4** (Convergence of the pooled law). *Let the calibration pairs be i.i.d. copies of  $(X, Y)$ , or a stationary ergodic sequence with that marginal law, and let  $\hat{G}_{\kappa,n}$  be the empirical distribution of the scores in stratum  $\kappa$ . Then  $\sup_w |\hat{G}_{\kappa,n}(w) - G_\kappa(w)| \rightarrow 0$  almost surely. If the smoothed estimate is KL-consistent, the excess log loss tends to the floor  $I(W; X | \kappa)$ .*

*Proof.* In the i.i.d. case the uniform convergence is the Glivenko–Cantelli theorem. In the stationary ergodic case the scores  $W_i = A(X_i, Y_i)$  are themselves stationary and ergodic, so the ergodic theorem applied to the indicators  $\mathbf{1}\{W_i \leq w\}$  gives almost sure convergence at each  $w$  in a countable dense set. Because distribution functions are monotone and  $G_\kappa$  is continuous, convergence on a dense set extends to uniform convergence. The second statement follows from Proposition 2, whose second term tends to zero under KL-consistency.  $\square$

Exchangeability alone does not give Lemma 4. An infinite exchangeable sequence is a mixture of i.i.d. sequences, and its empirical law converges to that of the random directing measure. Coverage therefore needs only exchangeability and holds at every sample size. The log-score statement is exact for every frozen pooled law (Proposition 2) and asymptotic only in identifying the law a conformal scheme ends up with.

**Corollary 5** (Gain of a conformal scheme over the unconditional law). *Under Proposition 2, let  $p_Y$  be the unconditional density of  $Y$  and suppose  $I(Y; X) < \infty$ . The conformal predictive gains over the unconditional law*

$$\mathbb{E} \log \hat{f}_X(Y) - \mathbb{E} \log p_Y(Y) = I(Y; X) - I(W; X | \kappa) - \mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa).$$

The gain is at most  $I(Y; X) - I(W; X | \kappa)$ , and the frozen score and stratification fix that bound. It is negative whenever  $I(W; X | \kappa) > I(Y; X)$ .

*Proof.* Add and subtract the oracle’s expected log score:

$$\mathbb{E} \log \hat{f}_X(Y) - \mathbb{E} \log p_Y(Y) = [\mathbb{E} \log p(Y | X) - \mathbb{E} \log p_Y(Y)] - [\mathbb{E} \log p(Y | X) - \mathbb{E} \log \hat{f}_X(Y)].$$

The first bracket is  $I(Y; X)$  by definition. The second bracket is  $L(\hat{F}) - H(Y | X)$ , which Proposition 2 evaluates as  $I(W; X | \kappa) + \mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa)$ . Substituting gives the display, and dropping the nonnegative KL term gives the bound.  $\square$

*Remark 6* (The calibration gain does not depend on the inputs). Calibration moves  $\hat{G}_\kappa$  toward  $G_\kappa$  and so removes the KL term. That term depends only on the pooled law of the score, and a forecaster whose inputs carry no information about the outcome gains it just as much. The part of the gain that depends on the inputs is  $I(Y; X) - I(W; X | \kappa)$ , and the frozen score sets it. For example, suppose  $Y$  is independent of  $X$  but the location model varies with  $x$ . The residual then depends on  $X$  through the model alone, and a scheme with exact coverage scores below the unconditional law by  $I(R; X)$ .

*Remark 7* (Distributional, localized and folded-score schemes). The input can also return through the score. Distributional conformal prediction (Chernozhukov et al., 2021) scores by an estimated conditional rank. Its floor is therefore  $I(W; X)$  for that rank, which is zero when the estimated conditional law is the true one. It closes the gap through the estimate and then stops. Density-based, histogram, localized and quantile constructions (Izbicki et al., 2020, 2022; Sesia and Romano, 2021; Guan, 2023; Romano et al., 2019) likewise lower the floor by what their conditioning retains, and calibration lowers it in none of them. The folded absolute residual can be read as a symmetric predictive with density  $h_{\text{sym}}(z) = \frac{1}{2}(p_R(z) + p_R(-z))$ . The same computation then adds  $\text{KL}(P_R \| H_{\text{sym}}) \leq \log 2$  for the discarded sign.

The floor measures how the law of the score varies with the input. Whether an input is typical or unusual under its own marginal law does not enter, and a rare input with ordinary noise adds nothing to  $I(W; X | \kappa)$ . Marginal coverage also does not say which inputs are over- or under-covered. We use *coverage* for the property of a set and *calibration* for the distributional property of a predictive.

We check Proposition 2, Corollary 5 and the folded case by grid integration in `check_gap.py`, computing each side independently. The checks include a distributional-conformal score from a misspecified model and a case where the gain is negative, and both sides agree to machine precision.

Stopping keeps the certificate, which concerns the marginal rank of the test score and is exact at its position. Stopping gives up  $I(W; X | \kappa)$ , the conditional information the frozen score discarded. A downstream model that uses the map’s output together with the input can recover part of that quantity.

## 4 The cost of stopping on 17,364 economic series

### 4.1 Moving the pooled law within a forecasting chain

To measure the cost of stopping we need a forecaster in which the empirical map can sit at more than one position, so we treat the map as a univariate transform. The raw empirical map is

a step function, so it is not invertible and has no density. We repair it by strictly monotone interpolation through its knots, with kernel-smoothed mass so that it can be log-scored. The result is an invertible, state-updated map of the observation coordinate, and it can sit anywhere in a chain of transforms.

We use the transform grammar of Cotton (2026). A forecaster is a *chain* of state-conditional invertible transforms, each removing one kind of structure and passing its residual on: a location tracker subtracts a running mean, a scale transform divides by a running deviation, a difference removes trend. A *terminal leaf* models the law of whatever residual remains, and the predictive in the original coordinate is the leaf’s law pushed back through the inverse chain, change of variables handled exactly. The base forecaster, **laplace**, runs roughly sixty candidate chains in parallel, combines their predictive densities with likelihood weights updated every observation, splices generalized Pareto tails onto the extremes, and scores its own probability integral transforms online. It ranks first in the public benchmark this study extends, and its own terminal leaf is tuned to CRPS.

We compare two configurations.

**stop.** Take the finished predictive, keep its centre, replace its shape with the pooled empirical law of the model’s own recent errors, kernel-smoothed. Quote that. Nothing acts downstream. This arm is a frozen conformal scheme of the covered type, so Proposition 2 applies with floor  $I(R; X)$ .

**compose-after.** Install the identical empirical object as the terminal leaf of *every* candidate chain. Writing  $R_t^c$  for chain  $c$ ’s residual and  $\hat{G}_t^c$  for the pooled empirical law of that chain’s own residual stream, the arm’s predictive is

$$\hat{F}_t = \sum_c w_t^c (T_t^c)^{-1}_{\#} \hat{G}_t^c,$$

where  $(T_t^c)^{-1}_{\#}$  pushes the map’s law back through chain  $c$ ’s inverse transform and the weights  $w_t^c$  are likelihood weights fitted online to the realized densities the maps assign, that is, a mixture model fitted on the maps’ output stream. Every predictive path passes through an empirical map, because the leaf override replaces the terminal leaf of every candidate chain. The pool weights therefore compare chains with each other and never compare the map with a parametric alternative. The one exception is the generalized Pareto tail splice, which replaces the density beyond a frozen tail region. It acts on the extremes of the composed predictive and is shared with the base arm. This arm carries no coverage certificate. It changes representation after the map instead of before it.

Both positions run with flat and with recency-weighted windows (exponential weights, half-life 150 observations), so staleness does not explain the differences.

## 4.2 Series, tiers and the null control

The corpus is FRED economic change series (differences of rates, production, employment and similar), non-price, with two tiers. The *breadth* tier uses 256 observations of history per forecast origin and every qualifying series: 17,364. The *deep* tier uses 1,024 observations and series with at least 1,100, so the 400-residual window is a strict subset of the history it could have used: about 6,400 series. Each series has twelve rolling origins, and we forecast at horizons 1, 6 and 12. Because origins within a series overlap, all intervals are bootstrap over series, not rows.

A control corpus of synthetic independent draws (Gaussian, Student- $t_3$ , Laplace) runs under the identical protocol at matched context. On the control there is no state to condition on, so what a constrained arm loses there is finite-sample overhead: the cost of estimating a 41-node empirical law rather than a parametric scale. We subtract that overhead from every headline number below and call the result *null-control-adjusted*. It is not the loss net of estimation, for two reasons. First, the estimation term  $\text{KL}(G \parallel \hat{G})$  depends on  $G$ , so its size on three simple iid laws gauges, but does not estimate, its size on economic residuals. Second, because the base forecaster adapts, its residuals are not exactly state-independent even on iid data. Without the adjustment the deficits are 0.51, 0.61 and 0.59 nats at horizons 1, 6 and 12 on the breadth tier (paired, 208,368 rows per horizon), so the adjustment removes about a fifth of the raw gap. A sharper control permutes each real series, which preserves the marginal residual law and destroys temporal state dependence. Where the two gauges disagree, the permutation control governs.

### 4.3 Cost of stopping and the share recovered by composing

On the 17,364 breadth-tier series, the null-control-adjusted log score deficits against the unconstrained base are as follows.

|                       | $h=1$ | $h=6$ | $h=12$ |
|-----------------------|-------|-------|--------|
| stop, total cost      | 0.397 | 0.475 | 0.462  |
| of which pooled shape | 0.267 | 0.315 | 0.275  |
| of which position     | 0.131 | 0.160 | 0.187  |

Raw per-arm deficits carry series-bootstrap 95% intervals of about  $\pm 0.02$  throughout. Composing recovers 33%, 34% and 40% of the cost at the three horizons. The deep tier gives raw stop costs of 0.34 to 0.40 nats with 40 to 55 percent recovered by the interior placement.

The deep-tier fraction is not a pure position effect. Between tiers three things change at once: the context grows from 256 to 1,024 observations, the series panels differ, and the 400-residual window truncates a 1,024-observation history while holding all of a 256-observation one. A stopped map need not use a 400-observation window, since it may retain every residual it has ever seen. A same-series, same-context ablation would separate these effects, with stop arms at windows 250, 400, 750 and all history and composed arms matched window for window on the ctx-1024 panel. Until it runs, we do not know whether the cost of stopping grows with how much the pooled window discards.

The arms are close on the median series. The mean cost is carried by a minority of series where stopping hurts badly.

Adaptivity does not substitute for position. Recency-weighting the stop arm’s window recovers 0.05 to 0.08 nats on the breadth tier, against 0.14 to 0.27 for moving the map’s position, raw in both cases. Proposition 2 dispatches level adaptation directly, since it acts on the pooled scores alone.

On CRPS the tiers disagree. On the deep tier the stopped arm degrades CRPS and the composed arm improves it, while on the breadth tier both positions sit within about 1% of the base. We do not use CRPS in any summary.

### 4.4 Empirical coverage of the stopped and composed arms

Conformal practice monitors coverage, so we measure it too. The series are economic time series and we assert no exchangeability, so this is empirical coverage and not the finite-sample certificate

of Section 1. The intervals are the 0.1 and 0.9 quantiles of the kernel-smoothed predictive, which need not coincide with the raw conformal order-statistic endpoints. The table therefore tests whether watching interval coverage would reveal the log score loss. It uses the breadth tier, paired rows and a target of 0.800.

| arm            | coverage | distance from target |
|----------------|----------|----------------------|
| unconstrained  | 0.870    | 0.070                |
| stop           | 0.784    | 0.016                |
| stop, weighted | 0.799    | 0.001                |
| compose-after  | 0.797    | 0.003                |

Every conformal arm sits nearer the nominal level than the unconstrained base, which overcovers at 0.870, so on coverage the conformal arms do better. On the log score the stopped arm loses 0.40 to 0.48 nats. Coverage tracks the rank of the outcome in the predictive, and the floor tracks conditional information the frozen score discarded. The two move independently, so monitoring coverage alone does not reveal the loss.

## 5 Discussion

The conformal recipe stops at its empirical map in the way training stops at an epoch, except that in learning the stop is chosen on validation evidence. The measured cost of the conformal stop is large, and composing an ordinary model past the map recovers a third to a half of it at no cost to empirical coverage. The conformal information gap identity locates the part of the loss that is fixed before calibration begins. When the floor is positive no calibration choice recovers any of it, and conformal calibration is most attractive exactly when a positive floor cannot be ruled out.

Because every calibration choice moves only the estimation term, a better calibrator cannot help. We recommend treating the empirical map as a coordinate change with a certificate attached at its position, and continuing to model on its output. The composed arm is one instance. In the sense of Remark 3 it is conformal prediction with the modelling finished, and its scored calibration stream measures whether the modelling was in fact finished.

**Limitations.** The count is a keyword screen with full-text verification of every label that could change the conclusion. The 599 stop labels rest on the screen and the composing criterion of Section 2, and the hand-coded random sample agrees with them. The prices are measured on one base forecaster and one economic corpus plus a synthetic control. The deep tier and the matched-window ablation of Section 4.3 are incomplete. The null-control adjustment gauges finite-sample overhead on simple iid laws and is not an unbiased estimate of the estimation term on economic residuals. Scoring floors the log density at  $-25$  per observation, which is not a proper score where it binds, and it binds unequally: on 1.78% of stopped-arm rows against 0.62% of base rows (breadth tier, paired). The inequality favours the stopped arm, and tightening the floor stepwise from  $-25$  to  $-10$  shrinks the raw stop deficit monotonically from 0.51 to 0.33 nats at horizon 1, so every gap reported here is a lower bound on its unclipped value. The composed arm has no coverage guarantee and is not proposed as a conformal method. The floor proposition covers invertible signed scores, and the absolute residual only in the symmetric reading of Remark 7. Quantile scores need their own treatment.

## A Proof of Proposition 2

Fix the fold, so  $(A, \kappa)$  is deterministic. Given  $X = x$ , change of variables gives the true density  $p(y|x) = p_W(A(x, y)|x) A'(x, y)$ , and the conformal density is  $\hat{f}_x(y) = \hat{g}_{\kappa(x)}(A(x, y)) A'(x, y)$  with  $\hat{g}_{\kappa}$  the density of the absolutely continuous estimate  $\hat{G}_{\kappa}$ . The Jacobian cancels in the log ratio, so

$$\text{KL}(P(\cdot|x) \parallel \hat{F}_x) = \text{KL}(P_W(\cdot|x) \parallel \hat{G}_{\kappa(x)}) = \text{KL}(P_W(\cdot|x) \parallel G_{\kappa(x)}) + \mathbb{E}_{W \sim P_W(\cdot|x)} \log \frac{g_{\kappa(x)}(W)}{\hat{g}_{\kappa(x)}(W)}.$$

Averaging over  $X$ , the first term is  $\mathbb{E}_X \text{KL}(P_W(\cdot|X) \parallel P_W(\cdot|\kappa(X))) = \text{I}(W; X | \kappa)$  by definition. Given  $\kappa(X) = k$  the score has law  $G_k$ , so the second term averages to

$$\mathbb{E}_{\kappa} \mathbb{E}_{W \sim G_{\kappa}} \log \frac{g_{\kappa}(W)}{\hat{g}_{\kappa}(W)} = \mathbb{E}_{\kappa} \text{KL}(G_{\kappa} \parallel \hat{G}_{\kappa}) \geq 0.$$

Every calibration operation is a function of the pooled scores alone, hence acts only on  $\hat{G}_{\kappa}$  and the second term.  $\square$

## B Reproduction

Every constant regenerates from a tidy store keyed (`task`, `source`, `dataset`, `uid`, `model`, `horizon`, `repr`, `metric`, `value`); the script `constants.py` beside this paper’s source performs the regeneration and fails if the manuscript disagrees with the store. Breadth tier rows are tagged `forecast:CONFB:ctx256:.../m6+hash`, deep tier `CONF:ctx1024`, controls `CONFNB/CONFNULL`; the hash covers the harness sources, the tracked diff, and the compiled core. The survey classifier, frame, and per-paper verdicts live beside the grammar paper. The hand-coded sample is `survey100.csv`. The exact checks of Section 3 are `check_gap.py` beside this paper. Study declarations and the fill protocol are in the `skaters` repository (`surrogate/bench/studies.py`, `skills/run-a-study.md`).

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